LEARNER GUIDE

Ratio, sharing and the unitary method

Percentages, Ratio, Growth and Bounds · Lesson 1. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A **ratio** compares amounts in **parts**. A smoothie made as fruit : yoghurt : milk = 2 : 1 : 1 has 4 equal parts. To **share** a batch in a ratio, find the size of **one part** first, then multiply. That is the **unitary method**: find one, then scale. Today the main activity is in the **Nutrition** panel, where the same idea reads food labels, scales recipes and shares the cost of shopping.

What you are learning to notice

- Total ÷ number of parts = one part. Every share follows from it.
- Order matters: 2 : 3 is not the same as 3 : 2.
- Multiplying or dividing every part by the same number gives the same mixture.
- Ratios work by multiplying, never by adding the same amount to each part.
- Food labels give amounts per 100 g, so different foods can be compared fairly.

Moving through it, one step at a time

Add the parts. Divide the total by the number of parts to find one part. Multiply by each ingredient's number of parts. Check that the shares add up to the total. For a label: find the amount in 10 g (divide by 10), then multiply by the number of 10 g lots in your portion.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Share a smoothie in the ratio 2 : 1 : 1

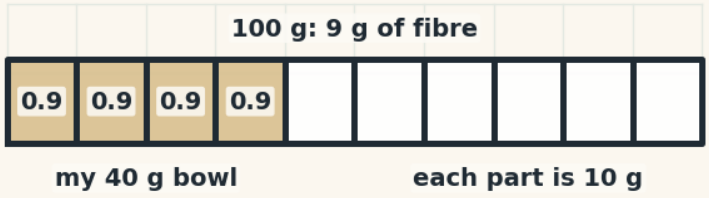
Make 600 ml of smoothie in the ratio fruit : yoghurt : milk = 2 : 1 : 1. How much of each?

1. Add the parts.	2 + 1 + 1 = 4 parts
2. One part	600 ÷ 4 = 150 ml
3. Fruit: 2 parts	2 × 150 = 300 ml
4. Yoghurt and milk: 1 part each	150 ml each
5. Check	300 + 150 + 150 = 600 ml

Fruit 300 ml, yoghurt 150 ml, milk 150 ml.

Example 2 — Read a label: per 100 g to my portion

Porridge oats have 9 g of fibre per 100 g. How much fibre is in a 40 g bowl?



1. 100 g is 10 lots of 10 g.	per 10 g: 9 ÷ 10 = 0.9 g
2. 40 g is 4 lots of 10 g.	4 × 0.9 = 3.6 g

A 40 g bowl has 3.6 g of fibre.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (highest common factors, a fraction of an amount, and money in pounds and pence), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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